Fiscal Year:	FY 2005	**Task Last Updated:**	FY 07/04/2005
PI Name:	Swain, Greg Ph.D.		
Project Title:	Advanced diamond electrodes for electrochemical based monitoring of spacecraft water		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Advanced environmental monitoring and control		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	48824	**Congressional District:**	8
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NOT AVAILABLE
Start Date:	08/26/2004	**End Date:**	08/31/2008
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Received NCE to 08/31/2008 per PI (9/2007)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Schultz, John R (Wyle Laboratory) Rutz, Jeffrey (Wyle Laboratory)		
Grant/Contract No.:	NNJ04HI10G		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The research is focused on developing advanced water quality monitoring methods for the potable and technical water supplies onboard the International Space Station. The methods being developed could also have broad application on other long-duration space flight and habitation projects. Electrochemical methods are being developed for water quality monitoring with two contaminant types targeted: (i) heavy metal ions including Ag(I), Cu(II), Cd(II), Pb(II) and Zn(II) and (ii) iodine and iodide as well as phenol and iodinated phenols. The key enabling material is the boron-doped diamond electrode, which provides a sensitive, reproducible and stable response for these analytes.		

Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The methods being developed could also provide sensitive, reproducible and stable detection of these, and other contaminants, in surface and ground water.
Task Progress:	Electrochemical methods for water quality monitoring are being pursued. The key enabling material is the boron-doped diamond electrode. This new electrode material meets the requirements for a wide range of applications and, in general, provides a sensitive, reproducible and stable response for chemical contaminants (those that are electroactive) in water supplies. We are focused on developing methods for the semicontinuous monitoring of two types of contaminants in the potable and technical water supplies on board the International Space Station: (i) heavy metal ions including Ag(I), Cu(II), Cd(II), Pb(II) and Zn(II) and (ii) iodine and iodide as well as phenol and iodinated phenols. This work builds upon the achievements of the Phase I effort (2003-2004) with regard to the monitoring of heavy metal ions, including the Ag(I) biocide, using anodic stripping voltammetry (ASV). It was demonstrated that the detection of heavy metal ions could be accomplished using a diamond electrode and differential pulse anodic stripping voltammetry (DPASV). The method was proven for the ground-based monitoring of heavy metal ions in contaminated soil, wastewater sewage sludge, drinking water and local river water. The first tasks in Year One of this current Phase II effort were to refine the method for on-orbit use in terms of adjusting the solution pH, testing the effect of dissolved oxygen, evaluating the electrode-to-electrode response reproducibility, evaluating the long-term stability of the electrode response, implementing pulsed galvanostatic deposition as a means for metal preconcentration in order to better control particle size and dispersion, evaluating the effect of intermetallic compound formation, and testing the effect of colloidal silver formation. Additionally, testing of ISS water simulants provided by the Wyle Laboratory (JSC) was goal. Significant progress was made on all of these tasks. First, it was demonstrated that increasing the solution pH from 4.5 to 5.2 (a value near that acceptable for on-board use) did not cause significant alteration in the detection figures of merit for five priority metal ions (Ag(I), Cu(II), Pb(II), Cd(II) and Zn(II)). It was also shown that the addition of Ca(II) and Mg(II) salts at concentrations identical to those presently used in the potable water caused less than a 4% change in the signal obtained for most of the priority metal ions tested. Second, elimination of the deoxygenation step (nitrogen purge) did not affect the signals for Ag(I) and Cd(II), but did cause a decrease in the signals for Pb(II) and Cu(II). Even with the decreased signals, though, a linear response was still observed for Pb(II) with a detection limit of at least 50 ppb. Third, the electrode-to-electrode response reproducibility for the stripping analysis of these priority metal ions was superb with a variability of less than 5%. Fourth, long-term testing of the electrode response stability for Ag(I) and Cd(II) was initiated. So far, the stripping signals signal for both have remained constant (RSD ~5%) over a period of several months of daily testing. Finally, the method was applied to the analysis of Ag(I) in simulated potable and technical water samples. Analysis of Ag(I) in the potable water sample yielded a result of 430 ± 2 ppb, which is within 2% of the expected value of 422 ppb. Analysis of Ag(I) in the technical water sample yielded a result of 530 ± 16 ppb, which is 9% higher than the expected value of 480 ppb. Work on the formation of interfering intermetallic compounds, pulsed deposition for minimizing these interferences, and the effects of colloidal Ag formation is currently in progress. We also performed some exploratory research to determine how useful the diamond electrode is for monitoring other priority chemical contaminants in water, in particular, iodine, iodide, phenols and iodinated phenols (I2 disinfection by-products). The phenolic by-products, 2-iodophenol, 4-iodophenol, and 2,4,6-triiodophenol, were targeted first. Cyclic voltammetry was used to study the oxidation reaction mechanism. Flow injection analysis with amperometric detection (FIA-EC) was employed to determine the detection figures of merit. Two types of diamond thin-film were evaluated: microcrystalline and nanocrystalline. Both diamond types yielded a sensitive, reproducible and stable response for the oxidative detection of 2-iodophenol and 4-iodophenol. For example, the linear dynamic range for the most common phenolic by-product, 2-iodophenol, was 5 orders of magnitude, the minimum concentration detected (S/N = 6) was 0.44 ppb, and the response precision was ca. 4%. A study of the oxidative detection of phenol was also completed. Cyclic voltammetry, FIA-EC, liquid chromatography with amperometric detection (LC-EC) and capillary electrophoresis with amperometric detection (CE-EC) were all used to quantify phenol levels in laboratory-prepared solutions. For example, the FIA-EC results demonstrated that diamond electrodes could be used to oxidatively detect phenol with a linear dynamic range from 28-280 ppb, a limit of detection of 28 ppb (S/N=3) and a response precision of 4 % (RSD). We are presently investigating the application of diamond electrodes for the detection of iodine and iodide levels in water using cyclic voltammetry and differential pulse voltammetry.
Bibliography Type:	Description: (Last Updated: 08/03/2009)
Articles in Peer-reviewed Journals	Sonthalia P, McGaw E, Show Y, Swain GM. "Metal ion analysis in contaminated water samples using anodic stripping voltammetry and a nanocrystalline diamond thin-film electrode." Anal Chim Acta 2004 Sep 20;522(1):35-44. http://dx.doi.org/10.1016/j.aca.2004.06.071 , Sep-2004
Papers from Meeting Proceedings	McGaw EA, Sonthalia P, Swain GM. "Diamond Thin-Film Electrodes for Monitoring Heavy Metal Ions in Water Supplies using Anodic STripping Voltammetry." International Conference on Environmental Systems, Rome, Italy, July 2005. SAE International (ICES, Rome, Italy), July 2005. , Jul-2005
Papers from Meeting Proceedings	Martin AN, Pearson LC, Swain GM. "Electrochemical Monitoring of Iodine Disinfection By-Products with Diamond Thin-Film Electrodes." International Conference on Environmental Systems, Rome, Italy, July 2005. SAE International (ICES, Rome, Italy), July 2005. , Jul-2005